Supplemental Material for "Finite-size decay-mode hybridization near a mean-field fold in an open Rydberg ensemble"

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I. GENERATOR, SYMMETRY SECTOR, AND PRECISION

We use $\gamma = 1$, $\Omega = 2$, and $V = 10$ unless stated otherwise. A permutation-invariant operator on N spin-1/2 particles has the form

$$\rho = \bigoplus_j \frac{\widetilde{\rho}_j}{d_j} \otimes I_{d_j}, \qquad d_j = \binom{N}{N/2-j} - \binom{N}{N/2-j-1}, \tag{S1}$$

where a binomial coefficient with lower argument -1 is zero. The stored PIQS blocks $\widetilde{\rho}_j$ sum over representation multiplicities. The trace is $\sum_j \operatorname{Tr} \widetilde{\rho}_j$, and the number of stored entries is $\sum_j (2j+1)^2 = \binom{N+3}{3}$. Local decay mixes different j blocks. All spectral and dynamical claims in the paper refer to this complete permutation-invariant operator space, not only $j = N/2$ and not to other permutation-symmetry sectors.

The Hamiltonian commutator is formed by column-major vectorization, $-i(I \otimes H - H^T \otimes I)$, followed by restriction to the Dicke blocks. The independent-emission coefficients follow the PIQS formulas [1], evaluated in double precision in `biorthogonal_generator.py`. For a target entry (j, m, m'), the diagonal decay coefficient is $-[N + (m + m')]/2$. Its off-diagonal sources have labels $(a, m_s, m'_s) = (j + \delta j, m + 1, m' + 1)$, with $\delta j = 0, +1, -1$. Their coefficients are, respectively,

$$\begin{aligned} &\frac{1}{2}\sqrt{(a+m_s)(a-m_s+1)(a+m'_s)(a-m'_s+1)}\,\frac{N/2+1}{a(a+1)}, \\ &\frac{1}{2}\sqrt{(a+m_s)(a+m_s-1)(a+m'_s)(a+m'_s-1)}\,\frac{N/2+a+1}{a(2a+1)}, \\ &\frac{1}{2}\sqrt{(a-m_s+1)(a-m_s+2)(a-m'_s+1)(a-m'_s+2)}\,\frac{N/2-a}{(a+1)(2a+1)}. \end{aligned} \tag{S2}$$

Only allowed source labels contribute; the first two expressions vanish when their denominators refer to an unavailable zero-spin transition. These coefficients are expressed in the multiplicity-summed storage convention.

An independent small-N check starts from the 2^N-dimensional spin Hilbert space with individual jump operators, and projects the full Liouvillian onto orthonormal symmetrized Pauli strings. The maximal matched spectral differences are shown in Table S1. This verifies the generator separately from eigensolver residuals, which cannot detect a common error in the matrix itself.

TABLE S1. Independent full-spin versus Dicke-block generator check. The invariance residual tests the full-spin action on the symmetrized Pauli basis.

N	Dimension	Maximum spectral difference	Invariance residual
1	4	1.99×10^{-15}	1.02×10^{-15}
2	10	1.01×10^{-14}	1.57×10^{-15}
3	20	3.77×10^{-14}	6.01×10^{-15}
4	35	4.46×10^{-14}	2.33×10^{-14}

This precision choice matters for a small gap. In the earlier PIQS-backed implementation used for exploratory calculations, internal emission coefficients led to a trace defect of approximately 4.7×10^{-7} at $N = 56$. The double-precision implementation reduces this to approximately 7.1×10^{-15} and changes that half-gap from 0.0003260261 to 0.0003309798. The finite-window gap exponent changes only from 0.140624 to 0.140501, but all figures retained in the present paper use the double-precision results. Earlier multi-drive quasipotential tests are not used as quantitative evidence in this manuscript.

II. MEAN-FIELD JACOBIAN AND INTERPRETATION OF THE FOLD

For the Bloch coordinates $q = (n, u, v)$ of the main text, the Jacobian is

$$J = \begin{pmatrix} -1 & 0 & -\Omega \\ Vv & -1/2 & -(\Delta - Vn) \\ \Omega - Vu & \Delta - Vn & -1/2 \end{pmatrix}. \tag{S3}$$

At stationarity, $v = -n/\Omega$ and $u = 2n(\Delta - Vn)/\Omega$. Substituting these relations into $\det(\lambda I - J)$, and reducing its difference from $\lambda[(\lambda+1)^2 + \Omega^2/2]$ by the polynomial constraints $F = F_n = 0$, gives zero identically. This is implemented symbolically in `pra_meanfield.py` and also checked numerically at both folds. For the parameters of the paper, the right fold is $(\Delta, n) = (4.5749637137, 0.4305200135)$; both folds have Jacobian eigenvalues $0, -1 \pm i\sqrt{2}$ to numerical precision.

The simple zero eigenvalue describes the linearization at the coalescing fixed points. The stationary cubic has a higher-order root at the cusp, but this does not change the algebraic multiplicity of the zero root in the displayed Jacobian polynomial. None of these identities identifies a finite-N spectral pair with a universal escape problem.

III. CONSTRUCTION AND VALIDATION OF PHYSICAL PREPARATIONS

Let a strictly positive single-atom density matrix have Bloch vector $\boldsymbol{b}$, with $r = |\boldsymbol{b}| < 1$. In the spin convention used by PIQS, the coordinates of the main text give

$$\boldsymbol{b} = (2u, -2v, 2n-1), \qquad \rho_1 = \sqrt{\det \rho_1}\, \exp[\operatorname{arctanh}(r)\, \widehat{\boldsymbol{b}} \cdot \boldsymbol{\sigma}]. \tag{S4}$$

The multiplicity-summed blocks of its product state are therefore

$$\widetilde{\rho}_j = d_j (\det \rho_1)^{N/2} \exp[2 \operatorname{arctanh}(r)\, \widehat{\boldsymbol{b}} \cdot \boldsymbol{J}_j]. \tag{S5}$$

We diagonalize the Hermitian exponent within each block and evaluate the scalar weights in logarithmic form. This avoids constructing a $2^N \times 2^N$ density matrix. The trace is checked before normalization.

For $N = 1, 2, 4$, the resulting excitation-number distribution agrees with $\binom{N}{k} n^k (1-n)^{N-k}$ to less than 1.7×10^{-16}, and the reconstructed single-atom coherences agree to less than 1.2×10^{-16}. Positivity follows directly from Eq. (S5); the single-atom states used in the calculations are strictly inside the Bloch ball. A separate $N = 1$–4 test starts from explicit tensor products and individual spin jumps, and integrates the full-spin equation with an adaptive DOP853 solver. At $t = 0, 0.25, 0.75, 1.5$, its excitation density and purity agree with Dicke-block exponential multiplication to less than 2×10^{-12}. This checks the preparation and physical trace convention in time evolution, independently of the spectral comparison in Table S1.

The high-branch difference is

$$\delta\rho_{\mathrm{hi}} = \rho_1(q_{\mathrm{hi}} + \eta e_{\mathrm{hi}})^{\otimes N} - \rho_1(q_{\mathrm{hi}} - \eta e_{\mathrm{hi}})^{\otimes N}, \tag{S6}$$

where q_{hi} is the surviving high mean-field fixed point, e_{hi} is its Euclidean-normalized real relaxation eigenvector with positive population component, and $\eta = 0.005$. The second difference is

$$\delta\rho_f = \rho_1(q_f)^{\otimes N} - \rho_{\mathrm{ss}}(\Delta), \qquad q_f = (0.15692704, 0.31861940, -0.07846352). \tag{S7}$$

The fold state is fixed while Δ is varied. It is not asserted to be a stationary or metastable state of the finite generator at those detunings.

For each detuning the real spectral pair is computed together with its transpose-left dual. If $H = W_d^T L R_d$ is the diabatic representation, diagonalizing $H = T\Lambda T^{-1}$ gives eigenmode bases $R = R_d T$ and $W = W_d (T^{-1})^T$. The observable residues are calculated from these exact eigenmode bases, not from the diagonal entries of the diabatic matrix. Scanning 15 detunings over $\Delta_*(32) \pm 0.06$ resolves the transfer of response; the endpoints and crossing are separately evaluated for $N = 40$. The largest right invariant-subspace residual in the $N = 32$ scan is 1.46×10^{-12}.

At $N = 32$, changing the ARPACK seed and shift from -0.76 to -0.80 changes the two real eigenvalues at the three diagnostic detunings by less than 9.0×10^{-15}. Halving η changes the residue vector divided by 2η by 4.96×10^{-6}, 7.29×10^{-6}, and 5.75×10^{-6} in relative Euclidean norm, respectively. These checks support the perturbative interpretation of the first preparation. They do not turn the second, finite displacement into a linearized mean-field trajectory.

IV. BIORTHOGONAL PROJECTION AND NORMALIZATION CONVENTIONS

The left eigenvectors are computed from L^T. In column-major vectorization the bilinear pairing $W^T R$ is the algebraic dual; physical observables are represented with the corresponding transpose. For a real decay mode of a Hermiticity-preserving generator, we phase-align and Hermitize the right and dual left eigenoperators, then enforce $W^T R = I$ in the selected two-dimensional subspace. Eigensolver, Hermiticity, trace, and subspace residuals are retained separately.

In this basis the derivative matrix $W^T L_1 R$ has two distinct real eigenvalues. Diagonalizing it defines the local diabatic modes. At the projection point, the derivative of the two-eigenvalue discriminant is unaffected by basis-derivative commutators. This gives the stationarity identity used in the main text. Numerically, the minimum is refined using

$$\delta\Delta = -\frac{e_a - e_b}{s_a - s_b} \tag{S8}$$

until $|\delta\Delta| < 10^{-11}$, with independently checked eigenmode and subspace residuals.

Let $P = R_d W_d^T$ be the generally oblique spectral projector at Δ_* and $Q = I - P$. Since P commutes with $L_* = L(\Delta_*)$, the exact spectral effective matrix away from that point is

$$\begin{aligned} H_{\mathrm{eff}}(z,\delta) &= H_P(\delta) \\ &\quad + \delta^2 W_d^T L_1 Q [z - Q(L_* + \delta L_1)Q]_Q^{-1} Q L_1 R_d. \end{aligned} \tag{S9}$$

The inverse acts on the range of Q. Thus the frozen linear matrix reproduces the value and first derivative at the reference point but not the complete neighboring spectrum. At $N = 40$, over $|\delta| \le 2g/|s_a - s_b|$, its largest tested half-gap relative error is 1.39×10^{-3}. A small absolute correction is not automatically negligible relative to a much smaller off-diagonal element.

In the stored convention of Eq. (S1), the physical trace norm of a Hermitian right operator is

$$\|R\|_1 = \sum_j \mathrm{Tr}\,|\widetilde{R}_j|. \tag{S10}$$

The corresponding norm of a dual observable is the maximum operator norm over its blocks. No extra representation-multiplicity factor is inserted on the left. Table S2 lists descriptive logarithmic slopes for the same data window under different normalizations. The balanced normalization uses $c_b/c_a = \sqrt{|V_{ba}/V_{ab}|}$ and forces equal magnitudes.

TABLE S2. Finite-window coupling exponents for $N = 24, 28, \ldots, 56$. The individual values are normalization-dependent.

Right-mode normalization	α_{ab}	α_{ba}	$(\alpha_{ab} + \alpha_{ba})/2$
Population peak	0.00963768	0.27136532	0.14050150
Physical trace norm	0.00356277	0.27744023	0.14050150
Balanced	0.14050150	0.14050150	0.14050150

V. BOUNDARY FLUX AND THE LIMITS OF TAIL ARGUMENTS

For a diagonal mask χ in the stored operator coordinates, define

$$J_{ab} = W_a^T [\chi, L] R_b, \qquad D_a = W_a^T \chi R_a, \qquad O_{ab} = W_a^T \chi R_b. \tag{S11}$$

The invariance equations $LR = RH$ and $W^T L = H W^T$ give

$$J_{ab} = (e_b - e_a) O_{ab} + V_{ab}(D_a - D_b). \tag{S12}$$

At the gap minimum, $e_a = e_b$, this is a flux representation of the cross coupling. The masks used in the diagnostic are

$$\chi_{(j,m,m')} = \mathbf{1}\left[\frac{k_m + k_{m'}}{2N} \le n_c\right], \qquad k_m = m + N/2. \tag{S13}$$

They are coordinate masks, not quantum channels or probabilities of measurement outcomes.

Eight cuts and nine sizes reconstruct the cross couplings with relative error below 1.6×10^{-8}. Nevertheless, splitting the weak flux into an absolute envelope and a cancellation factor is not invariant. For $n_c = 0.157$, their fitted exponents over $N = 24$–56 are approximately 0.13123 and 0.13942; at $n_c = 0.30$ they become 0.01137 and 0.25778. The denominator correction restores the same total coupling exponent. This prevents interpreting the two contributions at one selected cut as two independently determined activation actions.

The weak-coupling contraction can also be written $W_b^T(L - e_a)R_a$. In the exact projected subspace, $(L - e_a)R_a = V_{ba}R_b$, so the small quantity is transferred from a large cancelling sum into an exponentially small residual. An asymptotic calculation must determine that residual independently; using exact spectral modes to reconstruct it is not a WKB prediction.

VI. COMPLEX-DETUNING EXCEPTIONAL POINTS

We use a full-generator complex continuation $L(\zeta) = L_0 + \zeta L_1$, independently of the real spectral-parameter recursion. A two-sided block inverse iteration, with QR orthonormalization, gives right and transpose-left invariant subspaces. After dual normalization, define

$$H = W^T L R, \quad B = W^T L_1 R, \quad D = \left(\frac{H_{00} - H_{11}}{2}\right)^2 + H_{01}H_{10}. \tag{S14}$$

Because D is a similarity invariant, its derivative is

$$D' = \frac{(H_{00} - H_{11})(B_{00} - B_{11})}{2} + B_{01}H_{10} + H_{01}B_{10}. \tag{S15}$$

Newton iterations update both the detuning and invariant subspaces; they do not solve only a frozen two-by-two model. The QR basis remains regular at an EP, where individual eigenvectors become ill-conditioned.

TABLE S3. Double-precision complex-detuning EPs. $M = H - \frac{1}{2}\operatorname{Tr}(H)I$. Rates are in units of γ. The norm of M is evaluated with an orthonormal right-subspace basis.

N	ζ_{EP}	λ_{EP}	$\|M\|_F$	$\|M^2\|_F/\|M\|_F^2$
24	$3.136806063 + 0.026370766i$	$-0.751013173 + 0.031608734i$	0.25945	2.80×10^{-15}
32	$3.205686613 + 0.007464558i$	$-0.755104588 + 0.010634053i$	0.26380	2.10×10^{-15}
40	$3.250218904 + 0.002129005i$	$-0.757677321 + 0.003436485i$	0.26840	3.24×10^{-15}

The largest left or right subspace residual at the final EP points is 1.61×10^{-12}. Independent full-matrix ARPACK candidates have target-mode residuals below 1.68×10^{-13}. Direct eigenvalues at the EP still split by roughly 4–7×10^{-8} in floating-point arithmetic; we do not report this as an exact zero. The nonzero nilpotent part, simple zero of D, and the branch exchange provide complementary local evidence. Eight nearby candidate modes are retained; the nearest sampled external-mode distance is 0.457–0.460. This is not a global exclusion of other EPs or other spectral sectors.

For $N = 32$, we take $\zeta(\theta) = \zeta_{\mathrm{EP}} + re^{i\theta}$ with $r = 0.00223936747$, less than the distance to the conjugate EP. At 24 points per circuit, the one-circuit exchange and two-circuit return errors are 9.75×10^{-14} and 8.41×10^{-14}. The discriminant winds once and twice, respectively. Repeating with a different random seed, inverse-iteration shift, initial detuning, and 48 points per circuit changes the EP by 4.83×10^{-15} and common-node eigenvalues by at most 2.24×10^{-13}. A complex-direction central difference of D agrees with D' to relative error 4.10×10^{-8}.

VII. COHERENCE RECURSION, DERIVATIVES, AND REAL-ROOT VALIDATION

For real z, each denominator $X_q = zI - A_q - \Sigma_q(z)$ is inverted by a linear solve. Set $G_q = X_q^{-1}$ and $E_q = I - \Sigma_q'$. Differentiation of the positive-order recursion gives

$$\Sigma_{q-1}' = -U_{q-1}G_qE_qG_qD_{q-1}, \tag{S16}$$

$$\Sigma_{q-1}'' = U_{q-1}\left[2G_qE_qG_qE_qG_q + G_q\Sigma_q''G_q\right]D_{q-1}. \tag{S17}$$

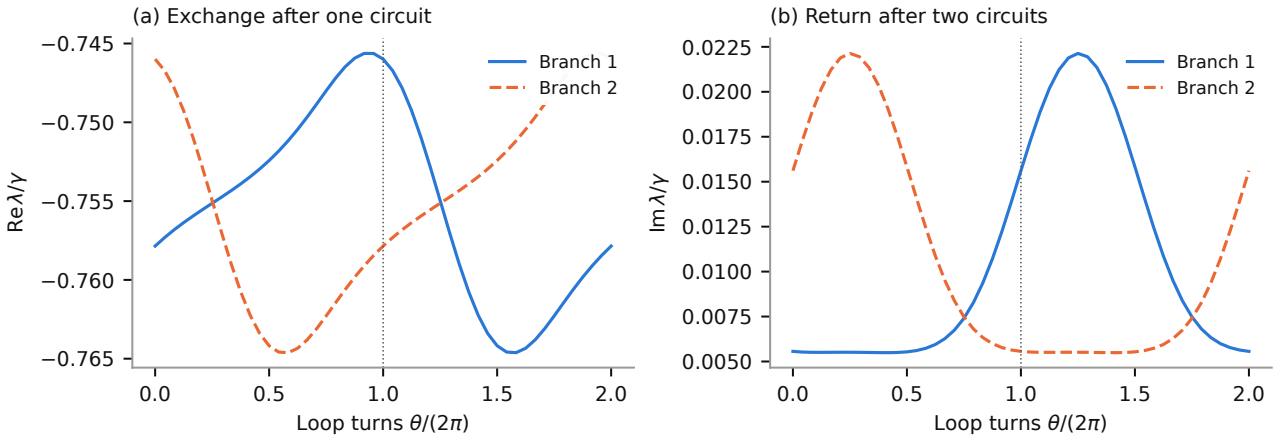


FIG. S1. Full-generator eigenvalue continuation around the $N = 32$ complex-detuning EP. Branch labels are followed continuously, rather than resorted at each point. The two branches exchange after one circuit and return after two. This complex-parameter loop is a spectral diagnostic, not a physical CPTP evolution.

With $K = A_0 + 2\,\mathrm{Re}\,\Sigma_0$, a simple real eigenbranch obeys

$$\mu' = l^T K' r, \qquad l^T r = 1, \tag{S18}$$

$$(K - \mu I)r' = -(K' - \mu' I)r, \qquad l^T r' = 0, \tag{S19}$$

$$\mu'' = l^T K'' r + 2 l^T K' r'. \tag{S20}$$

The middle equation is solved as a bordered linear system. All derivatives are with respect to the spectral variable z, not detuning. At the four minima $|f'| < 2.3 \times 10^{-12}$; eigenmode and differentiated-equation residuals are below 5.1×10^{-13}. For $N = 24$, central differences with step 10^{-4} agree with the first and second operator derivatives to relative errors 3.65×10^{-8} and 1.53×10^{-7}.

The root search selects the nearest effective real eigenvalue only within the local window. Such a selection is not a global branch-tracking algorithm and can jump between branches. A final residual check is therefore mandatory. In particular, $K = \mathrm{diag}(-0.80, -0.72)$ has no self-consistent root in $[-0.78, -0.74]$, although nearest-branch selection changes sign there and can make Brent's method report convergence. The implementation rejects this false root because the final residual is about 0.04. It also rejects a nearest effective eigenvalue with any nonzero imaginary part returned by the real-matrix eigensolver; a small complex pair is not rounded into a real root. Complex spectral inputs or detunings are rejected by the $2\,\mathrm{Re}\,\Sigma$ implementation.

For all eight retained roots, back-substitution reconstructs the complete right eigenoperator. The largest $\|LR - zR\|_2/\|R\|_2$ is 1.13×10^{-13} and the largest normalized trace is 8.2×10^{-14}. The $q = 0$ component carries between 0.360 and 0.419 of the full Euclidean norm. The smallest one-norm reciprocal condition estimates of the intermediate denominators at the roots range from 4.33×10^{-4} to 1.73×10^{-3}. These checks test the computed points, not absence of poles over an entire real or complex domain. Small residuals alone are not rigorous forward-error bounds for a nonnormal matrix; the agreement with independent full-generator spectra is also retained.

A. Truncating coherence orders

Keeping only $|q| \le q_{\max}$ defines a truncated matrix, not a CPTP approximation. At low cutoffs the nearest real pair can change identity; all candidate eigenvalues are saved. Representative relative half-gap errors are given in Table S4. Nonmonotonicity and different errors in the individual eigenvalues prevent treating one favorable gap value as convergence of the full spectrum.

VIII. TIME-DOMAIN COMPARISON AND ERROR DEFINITIONS

The full evolution is evaluated by sparse exponential multiplication for 121 times $t = 0, 0.1, \ldots, 12$, acting directly on the two initial differences. This avoids a time-discretized master-equation integrator. Trace conservation of the

TABLE S4. Relative half-gap error from coherence truncation. Full-generator values use all $|q| \le N$.

$q_{\max}$	$N = 40$	$N = 56$
8	$+5.1873$	$+165.17$
10	$+0.35523$	$+24.376$
12	$+0.025262$	$+3.7778$
14	-1.704×10^{-4}	$+0.65009$
16	-8.49×10^{-5}	-2.28×10^{-5}
20	-1.45×10^{-5}	$+2.35 \times 10^{-7}$
24	-5.66×10^{-7}	-1.83×10^{-6}

evolved differences is checked; the largest error in the three diagnostic detunings is below 9×10^{-16}.

The four-mode approximation includes the two real modes, the oscillatory eigenmode found near $-0.72 + 2.85i$, and its Hermitian-conjugate partner. Both right and transpose-left eigenmode residuals are checked. The residues are projected from the chosen preparations and observable, without fitting any time-domain parameters. For a lower time cutoff t_0, the reported relative error is

$$E(t_0) = \frac{\left[\sum_{t\in[t_0,12]} |\delta n_{\rm approx}(t) - \delta n_{\rm full}(t)|^2\right]^{1/2}}{\left[\sum_{t\in[t_0,12]} |\delta n_{\rm full}(t)|^2\right]^{1/2}}. \tag{S21}$$

The exponential envelope removed in the figure is not removed when calculating errors. Table S5 lists the $t_0 = 6$ comparisons. Earlier times generally contain larger contributions from omitted modes; for example, at the crossing the high-branch four-mode error is 1.49% over $4 \le t \le 12$.

TABLE S5. Relative L^2 errors over $6 \le t \le 12$ for $N = 32$. All values are percentages.

$\Delta - \Delta_*$	High, two real	High, four modes	Fold state, two real	Fold state, four modes
-0.06	25.14	0.481	46.34	5.14
0	25.10	0.506	69.47	5.00
$+0.06$	25.78	0.529	4.99	0.256

These are finite-time, preparation-dependent tests. The oscillatory pair has the less negative real part at the crossing, so it cannot be omitted solely because the target real pair has a small separation. Conversely, including four modes is not claimed to close the dynamics exactly or uniformly over all initial states.

IX. REPRODUCTION AND EVIDENCE SCOPE

The source bundle provides commands, scripts, JSON records, arrays, and a SHA256 manifest. The environment used Python 3.13.2, NumPy 2.2.6, SciPy 1.16.3, and QuTiP 5.3.1, with one linear-algebra thread. All calculations use `biorthogonal_generator.py`. The stored $N = 24$ coherence sample predates local geometric bracketing, and all four original root files predate the final branch-discontinuity guards. All eight roots were subsequently back-substituted with the final verifier; raw samples and source hashes preserve this provenance. Earlier quasipotential tests and unconverged instanton trajectories are not used as quantitative evidence here.

[1] N. Shammah, S. Ahmed, N. Lambert, S. De Liberato, and F. Nori, Open quantum systems with local and collective incoherent processes: Efficient numerical simulations using permutational invariance, Phys. Rev. A **98**, 063815 (2018).